RIPHAH
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Course: game Programming

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Project Title

Cricket League

Project Overview

Cricket League is a fast-paced 2D arcade cricket game. The Player swipes to hit cricket balls toward targets (boundaries, wickets, or specific zones). Bombs are replaced with “wide/Yorker traps” that reduce score or end the over. The goal is to score maximum runs within limited balls.

Project Objectives

Develop an engaging swipe-to-hit cricket gameplay.

Implement smooth ball trajectory and bat-swipe mechanics.

Design cricket-themed assets (bat, ball, pitch, stumps, crowd).

Apply game development concepts including modeling, texturing, level design, scripting, and rendering.

Deliver a fully functional Cricket League game within 2 weeks.

Research and Development Process

Research Phase

Studied Fruit Ninja swipe mechanics → adapt to cricket batting.

Analyzed cricket arcade games (stick cricket, batting tap games).

Researched ball physics, timing-based scoring, power-ups.

Explored swipe detection, particle effects for boundaries, and over limits.

Development Phase

Asset creation (bat, ball, stumps, pitch, crowd, traps).

Texture design (grass, ball shine, bat wood grain).

Asset importing and Unity project setup.

Level design (pitch background, boundary markers).

Scripting (swipe detection, ball spawn, score, over limit).

Testing, optimization, final rendering.

Sprint Planning (2 Weeks – 6 Sprints)

Sprint 1 – Modeling

Objective

Create all 3D/simple 2D game assets.

Tasks

Design cricket bat model.

Create cricket ball model.

Design stump and pitch elements.

Create “trap” assets (wide/red zone).

Prepare basic boundary markers.

Deliverable

Basic cricket game models completed.

Sprint 2 – Texturing

Objective

Apply textures to improve visual appearance.

Tasks

Apply wood texture to bat.

Add leather/red ball texture.

Texture pitch grass and stumps.

Design boundary rope texture.

Create trap warning textures.

Deliverable

All cricket assets textured.

Sprint 3 – Importing

Objective

Import and organize assets into Unity.

Tasks

Import bat, ball, pitch models.

Create prefabs (BallPrefab, TrapPrefab).

Organize folders (Scripts, Prefabs, Scenes, Audio).

Configure physics materials (bounce, drag).

Deliverable

All assets successfully imported.

Sprint 4 – Level Design

Objective

Design cricket gameplay scene.

Tasks

Create main cricket pitch scene.

Set up batting crease and bowler end.

Configure ball spawning zones.

Design scoreboard UI and over limit screen.

Add boundary hit markers.

Deliverable

Complete cricket level structure prepared.

Sprint 5 – Scripting

Objective

Implement core cricket mechanics.

Tasks

Develop ball spawning system (random line/length).

Implement swipe-to-bat mechanic (direction → runs).

Create run scoring system (4/6 runs for boundaries).

Add trap collision (wide/red zone reduces score).

Implement over completion (6 balls = game over).

Deliverable

Core gameplay fully functional.

Sprint 6 – Rendering & Testing

Objective

Finalize and optimize.

Tasks

Add particle effects for boundaries (sparks/dust).

Integrate crowd cheer sound effects.

Bug fixing (swipe misses, double counting).

Gameplay testing (timing, difficulty balance).

Optimize performance for low-end devices.

Deliverable

Final playable version of Cricket League.

Sprint Timeline

Session	**Sprint**	**Activity**
Class 1	Sprint 1	Modeling
Lab 1	Sprint 2	Texturing
Class 2	Sprint 3	Importing
Lab 2	Sprint 4	Level Design
Class 3	Sprint 5	Scripting
Class 4	Sprint 6	Rendering & Testing

Duration

Total Project Duration: 2 Weeks

Total Sprints: 6

Methodology: Agile Scrum

Expected Outcome

At the end of 2 weeks, Cricket League will be a fully functional swipe-based cricket arcade game with ball spawning, batting mechanics, run scoring, trap avoidance, and over-limited gameplay — mirroring the structure of Fruity Slash but with cricket theme.